

EXPERIMENT 9

9. AIM: Write a program for congestion control using leaky bucket algorithm

SOLUTION:

Leaky Bucket Algorithm

Step 1: Start.

Step 2: Read the **bucket size** and **output rate**.

Step 3: Read the **number of incoming packets**.

Step 4: Read the size of each packet one by one.

Step 5: Check whether the incoming packet can fit into the bucket.

- If **bucket + packet > bucket_size**, **discard the packet** due to overflow.
- Otherwise, **add the packet to the bucket**.

Step 6: Transmit data from the bucket at the **fixed output rate**.

Step 7: Subtract the transmitted amount from the bucket.

Step 8: Repeat Steps 4–7 for all incoming packets.

Step 9: After all packets are processed, transmit the **remaining data** in the bucket at the fixed output rate.

Step 10: Stop.

Source Code:

```
#include <stdio.h>
```

```
int main()
{
    int bucket_size, output_rate;
    int n, packet;
    int bucket = 0;
    int transmitted;

    printf("Enter bucket size: ");
    scanf("%d", &bucket_size);

    printf("Enter output rate: ");
    scanf("%d", &output_rate);

    printf("Enter number of packets:\n");
    scanf("%d", &n);

    for (int i = 1; i <= n; i++)
    {
        printf("\nEnter size of packet %d:\n", i);
        scanf("%d", &packet);

        /* Check for bucket overflow */
        if (bucket + packet > bucket_size)
        {
            printf("Packet %d discarded - Bucket Overflow\n", i);
        }
        else
        {
            bucket = bucket + packet;
            printf("Packet %d added to bucket\n", i);
        }
    }
}
```

```

    /* Transmit at fixed output rate */
    if (bucket > 0)
    {
        if (bucket >= output_rate)
            transmitted = output_rate;
        else
            transmitted = bucket;

        bucket = bucket - transmitted;

        printf("Transmitted = %d\n", transmitted);
        printf("Remaining in bucket = %d\n", bucket);
    }
    else
    {
        printf("Bucket is empty\n");
    }
}

/* Transmit remaining data */
while (bucket > 0)
{
    if (bucket >= output_rate)
        transmitted = output_rate;
    else
        transmitted = bucket;

    bucket = bucket - transmitted;

    printf("\nTransmitted = %d\n", transmitted);
    printf("Remaining in bucket = %d\n", bucket);
}

return 0;
}

```

Output:

Enter bucket size: 10
Enter output rate: 4
Enter number of packets:
5

Enter size of packet 1:
6
Packet 1 added to bucket
Transmitted = 4
Remaining in bucket = 2

Enter size of packet 2:
8
Packet 2 added to bucket
Transmitted = 4
Remaining in bucket = 6

Enter size of packet 3:

3

Packet 3 added to bucket

Transmitted = 4

Remaining in bucket = 5

Enter size of packet 4:

9

Packet 4 discarded - Bucket Overflow

Transmitted = 4

Remaining in bucket = 1

Enter size of packet 5:

2

Packet 5 added to bucket

Transmitted = 3

Remaining in bucket = 0